Arunabha Majumder

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Portfolio: arunabha22.github.io/arunabha-majumder

Isar Aerospace SE, Loads & Dynamics team, Parsdorf

Working Student, Launch Vehicle Structures

Dear Isar Aerospace team,

I'd like to apply for the Working Student position in the Loads & Dynamics team. I am a PhD student in mechatronics and robotics at Aalborg University, and comparing models with test data is a big part of what I already do, so this role feels like a natural next step into space.

In my research I designed and modelled a novel 5-bar planar parallel robot with variable stiffness mechanisms (VSM), estimated its stiffness with less than 10% error and quantified human arm impedance. Getting there meant testing it on real hardware and comparing what my model predicted with what we measured, until it matched. For the arm impedance study with this robot (6 participants), Qualisys cameras recorded the arm position data, and I implemented an Arduino-based trigger so the robot's motion and the recording stayed in sync. In a separate study I collected EMG data from 12 participants wearing my exoskeleton, using the Qualisys data acquisition system, so I know how much careful sensor setup and documentation matter.

I also designed a hybrid shoulder exoskeleton with a variable stiffness mechanism and a 6 Nm motor, integrated the motor, encoder and force/torque sensors, and measured its stiffness on a physical test setup. It uses over 25% less motor energy and won the Best Paper Award at ISRM 2026. In my Master's I designed the outer PID loop, with integral anti-windup and error nonlinearisation, of a neural network-based gain-scheduled controller in MATLAB/Simulink. I also hold a Senior Diploma in Fine Arts and do graphic design (Illustrator, Photoshop, Procreate), which helps a lot when turning data into clear plots and figures.

I'm ready to move to the Munich area and I'm willing to learn German. I would love to bring new and innovative ideas to your data workflows and tools, and to learn as much as I can about loads and dynamics from your team. My portfolio, with videos and results from my work, is at arunabha22.github.io/arunabha-majumder. Thank you for reading.

Best,
Arunabha Majumder